

C616

Middle School Science: Content Knowledge

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A sixth-grade teacher who has taught for nine years hands students a 60-gram wooden block with a volume of 75 cm^3 . The block came from a local furniture maker. Will it float in water?

- A. No; 60 grams is too heavy to float.
- B. No; its density is 1.25 g/cm^3 .
- C. Yes; its density is 0.8 g/cm^3 , less than water's.
- D. Yes, because all wooden objects float.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A sample has a density of 2.5 g/cm^3 and a volume of 4 cm^3 . What is its mass?

- A. 6.5 g
- B. 1.6 g
- C. 10 g
- D. 0.625 g

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Students at a school with a new science lab observe four changes during a week when the lab's heater was broken. Which is a chemical change?

- A. Sugar dissolving in water
- B. Ice melting on the counter
- C. An iron nail rusting
- D. Water boiling, because bubbles always mean a new substance

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A student whose parents are both chemists puts 50 g of baking soda and vinegar, including the bag, in a sealed zip bag. The bag puffs up as gas forms. The experiment is filmed for the school news. What is the mass of the sealed bag afterward?

- A. 50 g
- B. It cannot be predicted without a balanced equation
- C. More than 50 g, because a gas was made
- D. Less than 50 g, because gas escaped

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

In an open beaker, 100 g of reactants produce products that read 96 g on the scale. What is the best explanation?

- A. 4 g of matter was destroyed.
- B. The scale must be broken, since mass never changes on a scale.
- C. Products always weigh less than reactants.
- D. A gas formed and escaped into the air.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

At a family restaurant known for its chili, a metal spoon left in a hot bowl becomes hot at the handle. The restaurant seats 80 people. How was thermal energy mainly transferred to the handle?

- A. Cold moved from the spoon into the soup
- B. Conduction
- C. Radiation
- D. Convection

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

In July, the Northern Hemisphere has summer even though Earth is near its farthest point from the Sun. A science club that meets in the library notes this in its newsletter. What causes summer in the Northern Hemisphere?

- A. Earth is closest to the Sun in July.
- B. The Sun gives off more energy in summer.
- C. The Southern Hemisphere is tilted toward the Sun.
- D. The Northern Hemisphere is tilted toward the Sun, getting more direct rays and longer days.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

When it is summer in the Northern Hemisphere, what season is it in Australia?

- A. Winter
- B. Summer, because seasons are the same worldwide
- C. No season, because Australia is on the equator
- D. Spring

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A teacher who hosts a monthly star party at a county park asks why the Moon looks like a crescent on some nights. Attendance averages 40 people. What causes Moon phases?

- A. The Moon makes less of its own light at times.
- B. Clouds block part of the Moon.
- C. Earth's shadow covers part of the Moon.
- D. We see different amounts of the Moon's sunlit half as it orbits Earth.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

During a full moon, where is the Moon relative to Earth and the Sun?

- A. At a right angle to the Earth-Sun line
- B. On the opposite side of Earth from the Sun
- C. Between Earth and the Sun
- D. Exactly in line, causing an eclipse every month

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A dog that won a regional agility contest has a scar on its ear and brown eyes. Its owner trains it every morning. Which trait could the dog pass to its puppies?

- A.** All of its traits, since puppies copy their parents
- B.** Brown eye color
- C.** The scar on its ear
- D.** The agility skills it learned

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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12.

A cyclist on a team sponsored by a bike shop rides 30 km in 1.5 hours on a windy day. What is her average speed?

- A.** 20 km/h
- B.** 30 km/h
- C.** 0.05 h/km
- D.** 45 km/h

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A sixth-grade teacher ~~who has taught for nine years~~ hands students a 60-gram wooden block with a volume of 75 cm³. ~~The block came from a local furniture maker.~~ Will it float in water?

Target word: float or sink **Decides:** Density = $60 \div 75 = 0.8 \text{ g/cm}^3$.

A	Camouflage	Total mass alone does not decide floating.
B	Inverted operation	Divides volume by mass.
C	Correct	Less dense than 1 g/cm ³ floats.
D	Absolute language	Some very dense woods sink; density decides.

2. Answer C Concept 1 • transfer

A sample has a density of 2.5 g/cm³ and a volume of 4 cm³. What is its mass?

Target word: solve for mass **Decides:** $m = d \times V$.

A	Adjacent category	Adds the values instead of multiplying.
B	Inverted operation	Divides volume by density.
C	Correct	$2.5 \times 4 = 10$.
D	Confusion set	Divides density by volume.

3. Answer C Concept 2

Students at a school ~~with a new science lab~~ observe four changes ~~during a week when the lab's heater was broken.~~ Which is a chemical change?

Target word: identify the chemical change **Decides:** A new substance forms.

A	Adjacent category	A mixture forms; the sugar can be recovered.
B	Confusion set	A change of state only.
C	Correct	Iron combines with oxygen to form a new substance, iron oxide.
D	Absolute language	Boiling bubbles are water vapor, the same substance.

4. Answer A Concept 3

A student ~~whose parents are both chemists~~ puts 50 g of baking soda and vinegar, including the bag, in a sealed zip bag. The bag puffs up as gas forms. ~~The experiment is filmed for the school news.~~ What is the mass of the sealed bag afterward?

Target word: mass after reaction in closed system **Decides:** Sealed: nothing enters or leaves.

A	Correct	Mass is conserved in a closed system.
B	Adjacent category	Conservation of mass holds for any reaction.
C	Inverted operation	Gas is made from the reactants, not added.
D	Dropped element	Ignores that the bag is sealed.

5. Answer D Concept 3 • transfer

In an open beaker, 100 g of reactants produce products that read 96 g on the scale. What is the best explanation?

Target word: reverse: explain apparent mass loss **Decides:** Open system.

A	Inverted operation	Matter is not destroyed in chemical reactions.
B	Absolute language	In an open system, measured mass can drop.
C	Confusion set	Total mass of products equals reactants when all are captured.
D	Correct	Mass was conserved but left the beaker.

6. Answer B Concept 4

At a family restaurant ~~known for its chili~~, a metal spoon left in a hot bowl becomes hot at the handle. ~~The restaurant seats 80 people.~~ How was thermal energy mainly transferred to the handle?

Target word: method of heat transfer **Decides:** Direct contact through a solid.

A	Inverted operation	Thermal energy flows from warm to cool, not "cold" flowing.
B	Correct	Energy passes particle to particle through the metal.
C	Adjacent category	Energy by waves, not the main path through the spoon.
D	Confusion set	Needs fluid flow; the spoon is a solid.

7. Answer D Concept 5

In July, the Northern Hemisphere has summer even though Earth is near its farthest point from the Sun. A science club ~~that meets in the library~~ notes this in its newsletter. What causes summer in the Northern Hemisphere?

Target word: cause of seasons **Decides:** Axial tilt.

A	Confusion set	The stem states the opposite; distance does not cause seasons.
B	Adjacent category	The Sun's output is nearly constant.
C	Inverted operation	That is the arrangement in January.
D	Correct	Tilt changes the angle and duration of sunlight.

8. Answer A Concept 5 • transfer

When it is summer in the Northern Hemisphere, what season is it in Australia?

Target word: reverse hemisphere **Decides:** Tilt favors one hemisphere at a time.

A	Correct	The Southern Hemisphere is tilted away from the Sun.
B	Absolute language	Hemispheres have opposite seasons.
C	Confusion set	Most of Australia is well south of the equator.
D	Adjacent category	Spring and fall occur between the extremes.

9. Answer D Concept 6

A teacher ~~who hosts a monthly star party at a county park~~ asks why the Moon looks like a crescent on some nights. ~~Attendance averages 40 people.~~ What causes Moon phases?

Target word: cause of phases **Decides:** Viewing angle of the sunlit half.

A	Dropped element	The Moon makes no light; it reflects sunlight.
B	Adjacent category	Weather does not create a regular cycle of phases.
C	Confusion set	That describes a lunar eclipse.
D	Correct	The Sun always lights half; our view changes.

10. Answer B Concept 6 · transfer

During a full moon, where is the Moon relative to Earth and the Sun?

Target word: reverse: position from phase **Decides:** Full sunlit face toward Earth.

A	Adjacent category	That gives a quarter moon.
B	Correct	Earth is roughly between the Sun and Moon.
C	Inverted operation	That is the new moon position.
D	Absolute language	The Moon's tilted orbit means eclipses are rare.

11. Answer B Concept 7

A dog ~~that won a regional agility contest~~ has a scar on its ear and brown eyes. ~~Its owner trains it every morning.~~ Which trait could the dog pass to its puppies?

Target word: inherited trait **Decides:** Genes code for it.

A	Absolute language	Only genetic traits are inherited.
B	Correct	Eye color is inherited through genes.
C	Confusion set	An injury is acquired during life.
D	Inverted operation	Learned skills are acquired, not inherited.

12. Answer A Concept 8

A cyclist ~~on a team sponsored by a bike shop~~ rides 30 km in 1.5 hours ~~on a windy day~~. What is her average speed?

Target word: average speed **Decides:** Distance ÷ time.

A	Correct	$30 \div 1.5 = 20$.
B	Wrong stem number	Reports the distance as the speed.
C	Confusion set	Divides time by distance.
D	Inverted operation	Multiplies distance by time.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Density decides floating	Density = mass ÷ volume; an object floats in water if its density is less than about 1 g/cm ³ , regardless of its total mass.	Thinking heavy objects sink and light ones float.	A mass and a volume given together.
2	Physical vs. chemical change	A chemical change makes a new substance (clues: gas, color change, heat, or precipitate that is not just a state change); a physical change alters form or state only.	Treating melting, boiling, or dissolving as chemical changes.	"Which is a chemical change?"
3	Conservation of mass	In a closed system total mass stays the same through any change; apparent mass loss in an open container means gas escaped.	Believing mass is destroyed or created when a gas is produced.	A sealed bag vs. an open beaker.
4	Heat transfer	Thermal energy moves from warmer to cooler matter by conduction (contact), convection (fluid flow), or radiation (waves).	Thinking cold flows into warm objects or mixing up the three methods.	A spoon, a pot of water, or sunlight.
5	Cause of the seasons	Seasons come from Earth's 23.5° axial tilt, which changes how directly sunlight strikes and how long days last; distance to the Sun is not the cause.	Believing Earth is closer to the Sun in summer.	Hemisphere, month, or "why is it warmer?"
6	Moon phases	Phases show how much of the Moon's sunlit half faces Earth as the Moon orbits; Earth's shadow causes eclipses, not phases.	Blaming Earth's shadow for phases.	"Why does the Moon's shape change?"
7	Inherited vs. acquired traits	Only traits coded in genes pass to offspring; traits gained through experience or injury are not inherited.	Believing learned skills or built muscles pass to offspring.	A trait gained during an organism's life.
8	Average speed	Average speed = total distance ÷ total time.	Multiplying distance by time or reporting the distance.	A distance and a time for a trip.

Course scope (public description): This course covers the content knowledge that a middle-level science teacher is expected to know and understand. Topics include scientific methodologies, history of science, basic science principles, physical sciences, life sciences, earth and space sciences, and the role of science and technology and their impact on society.